Fiscal Year:	FY 2014	**Task Last Updated:**	FY 07/10/2013
PI Name:	Beckermann, Christoph Ph.D.		
Project Title:	Effect of Convection on Columnar-to-Equiaxed Transition in Alloy Solidification		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	10/01/2010	**End Date:**	02/28/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 2/28/2014 per NSSC information; continuation grant awarded (Ed., 7/14/14) NOTE: End date is 9/30/2014 per NSSC info (Ed., 7/3/13)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	The project examines the mechanisms giving rise to the columnar-to-equiaxed grain structure transition (CET) during alloy solidification. On earth, experimental investigations of the CET are affected by thermosolutal buoyant convection and grain sedimentation/flotation, making it impossible to separate these effects from the effects of solidification shrinkage and diffusive processes in determining mechanisms for the CET. Long duration microgravity experiments suppress the convective effects and grain movement, thus isolating the shrinkage and diffusive phenomena. The project increases the base of knowledge relevant to the development of solidification microstructure/grain structure of metals and alloys. Therefore, this topic is of high interest from a fundamental science point of view and it is important to those engineers practicing casting and other solidification processes. Open scientific questions include the role played by melt convection, fragmentation of dendrite arms, and the transport of fragments and equiaxed crystals in the melt. The research utilizes computational models at three different length scales: phase-field, mesoscopic, and volume-averaged models. The phase-field model is needed to resolve the growth and transport processes at the scale of the microstructure, the mesoscopic model allows for simulations at the scale of individual grains, while the volume-averaged model is used to perform simulations of entire experiments. The models help to define and interpret previous and future microgravity and ground-based experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The columnar-to-equiaxed transition (CET) in the grain structure of metal alloy castings has fascinated researchers in the solidification area for more than 50 years. The CET refers to the transition between the elongated grains in the outer portions of a casting and the more rounded grains in the center. Understanding this transition is fundamental to determining what type of grain structure forms in castings of most metal alloys (steel, aluminum, copper, etc.). Often, a fully equiaxed structure is preferred, but the fully columnar structures of many turbine blades are an important exception. In addition to its high practical significance, the CET represents a “holy grail” in the area of modeling and simulation of casting. This is because in order to realistically predict the CET, almost every physical phenomenon at every length scale must be taken into account simultaneously: heat transfer, solute transport, melt flow, and the transport of small dendrite fragments and equiaxed grains on the scale of the casting; the thermal/solutal/mechanical interactions between the growing grains/dendrites; and the nucleation of grains (especially in the presence of grain refiners) and fragmentation of existing dendrites. The research will not only provide an improved understanding of the CET, but also models and computer simulations of the grain structure formation in metal castings that can be used by industry to better understand and optimize their casting processes.
Task Progress:	Progress was made in all areas of the research on the columnar-to-equiaxed transition (CET) in alloy solidification: phase-field simulation, mesoscopic simulation, and macroscopic simulation. A Science Requirement Document for future CET microgravity experiments was completed and the PI passed a Science Concept Review that was held at NASA MSFC in March 2013. Three-dimensional phase-field simulations of alloy solidification are being conducted to study the dendrite fragmentation process on a microscopic (microstructure) scale. Fragmented dendrite sidebranches are believed to be a potent source of equiaxed grains. For this purpose, simulations are being conducted for columnar dendritic growth with an imposed temperature gradient and cooling rate. After a fully dendritic structure is obtained, the cooling rate is suddenly reduced. This leads to fragmentation of the dendrites at the junction between primary and secondary sidebranches. The fragmentation dynamics and rates are being studied as a function of the growth conditions. Considerable effort was devoted to parallelizing the code in order to allow for large scale simulations to be conducted. Speedups by more than a factor of 50 have been achieved. In addition, the deformation of dendrites is being studied using phase-field simulations. Such deformation can also lead to fragmentation. During the next project year, these simulations will be continued. A theory of fragmentation will be developed. Mesoscopic simulations of columnar and equiaxed solidification are being performed in order to investigate in detail the evolution of the grain structure on an intermediate scale. In this type of simulation, the evolution of the dendrite envelopes is tracked, while the solute field is calculated only in the extra-dendritic space between the envelopes. A three-dimensional computer code has been written and preliminary simulations have been performed to compare the predicted envelope shapes with available measurements. The next step is to include melt convection. This work is well underway. Macroscopic simulations are being conducted to study the CET on the scale of an entire casting. A previously developed volume-averaged model is used for these simulations. The governing equations are solved using the public domain OpenFoam CFD software platform. The code was tested for columnar and equiaxed solidification without melt convection and transport of solid. Gravity-driven convection and grain sedimentation/floatation are currently being added to the model. Macroscopic simulations will be conducted to help analyze and design future microgravity experiments. A Science Requirement Document was prepared for microgravity CET experiments on the ISS. The hypothesis of the experiments is that fragmentation of columnar dendrites can lead to a CET in non-grain-refined alloys. As opposed to current theories, the experiments will show that a growth velocity increase and nucleation of equiaxed grains ahead of the columnar front are not necessary conditions for a CET to occur. It is proposed to conduct solidification experiments where a CET is induced by a growth velocity decrease, rather than an increase. The Low Gradient Furnace (LGF) on the MSL will be used for the proposed experiments. Experiments are proposed for various Al-Cu and Al-Si alloys. All CET experiments will consist of two solidification phases (which are preceded by a homogenization phase). During the first phase, a columnar grain structure will be established by imposing a certain temperature gradient and cooling rate (furnace translation rate). During the second phase, the portion of the sample that is still in mushy and liquid states should be located in the hot zone of the furnace and be held adiabatically, while all heat extraction occurs from the cold zone through the already solidified portion. No further furnace translation is desired. The imposition of adiabatic conditions along the sample that is still in a fully or partially liquid state will prevent additional nucleation of solid. During the second phase, the columnar front velocity will first experience a growth velocity increase until all remaining melt superheat is dissipated. Then, the columnar front will slow down because it grows into an isothermal undercooled liquid. This growth velocity decrease will induce fragmentation of the columnar grains. Because the entire melt is undercooled, the fragments will not re-melt and a CET can occur. In March 2013, the PI passed a Science Concept Review that was held at NASA MSFC. ED. NOTE (7/14/2014): Project continued by "Effect of Convection on Columnar-to-Equiaxed Transition in Alloy Solidification--NNX14AD69G," grant # NNX14AD69G with period of performance 3/1/20014-2/29/2016. See that project for subsequent reporting.
Bibliography Type:	Description: (Last Updated: 12/04/2024)

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